Observability:

The state $x_0 \neq 0$ is said to be unobservable if, given $x(0) = x_0$, and $u(\tau) = 0$ for $\tau \geq 0$, then $y(\tau) = 0$ for $\tau \geq 0$.

The system is said to be completely observable if there exists no nonzero initial state that is unobservable.

Test of observability

give $\dot{x} = Ax + Bu, \quad x(0) = x_0$

$y = Cx + Du$

the observability matrix:

$$J_o[A, C] = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}$$

The system is completely observable if and only if $J_o[A, C]$ has full column rank n.

Detectability:

A plant is said to be detectable if its unobservable subspace is stable

Observer canonical Form

$$A_o = \begin{bmatrix} -a_1 & 1 & 0 & \cdots & \\ -a_2 & 0 & 1 & & \vdots \\ \vdots & & \ddots & \ddots & 1 & 1 \\ -a_n & & & & 0 & 0 \end{bmatrix}; \quad B_o = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix};$$

$$C_o = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \end{bmatrix}$$

e.g. $G(s) = \dfrac{5}{s^2+3s+12}$

$$\Rightarrow A_o = \begin{bmatrix} -3 & 1 \\ -12 & 0 \end{bmatrix} \qquad B_o = \begin{bmatrix} 0 \\ 5 \end{bmatrix}$$

$$C_o = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -3 & 1 \\ -12 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 5 \end{bmatrix}$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$